INCH-POUND

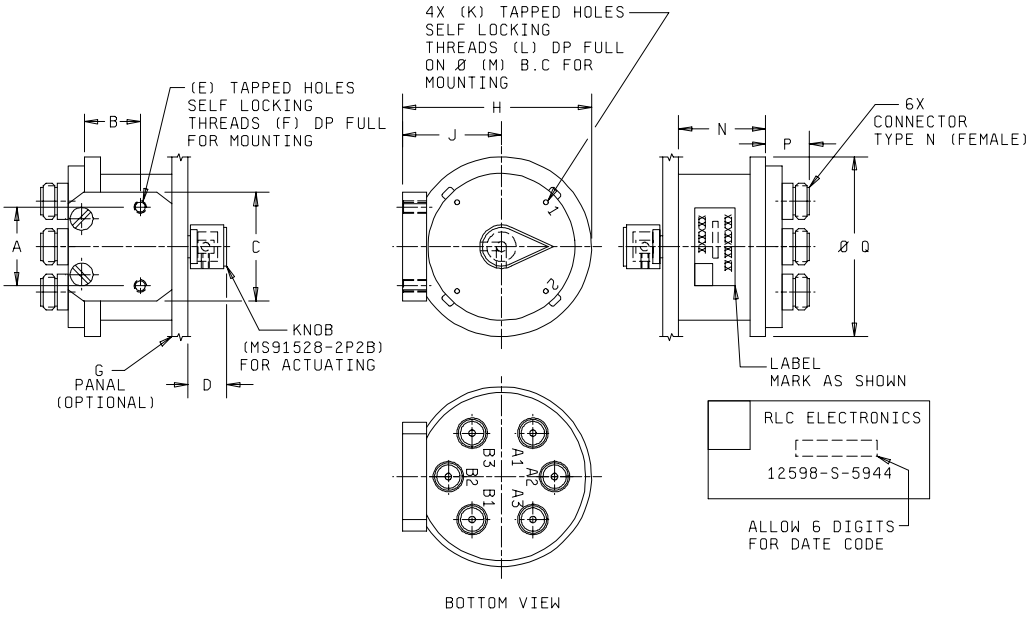
MIL-DTL-3928/9D
5 July 2001
SUPERSEDING
MIL-S-3928/9C
15 July 1988

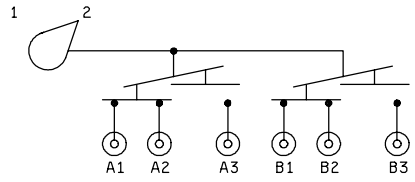
DETAIL SPECIFICATION SHEET

SWITCHES, RADIO-FREQUENCY
TRANSMISSION LINE (COAXIAL) (MANUALLY AND MOTOR OPERATED)
CLASS 1 (TYPE N CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the switch described herein shall consist of this specification and MIL-DTL-3928.




POSITION 1 CONNECTS A2 TO A3 | B2 TO B3
POSITION 2 CONNECTS A1 TO A2 | B1 TO B2

FIGURE 1. Dimensions and configuration for PIN M3928/9-04.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

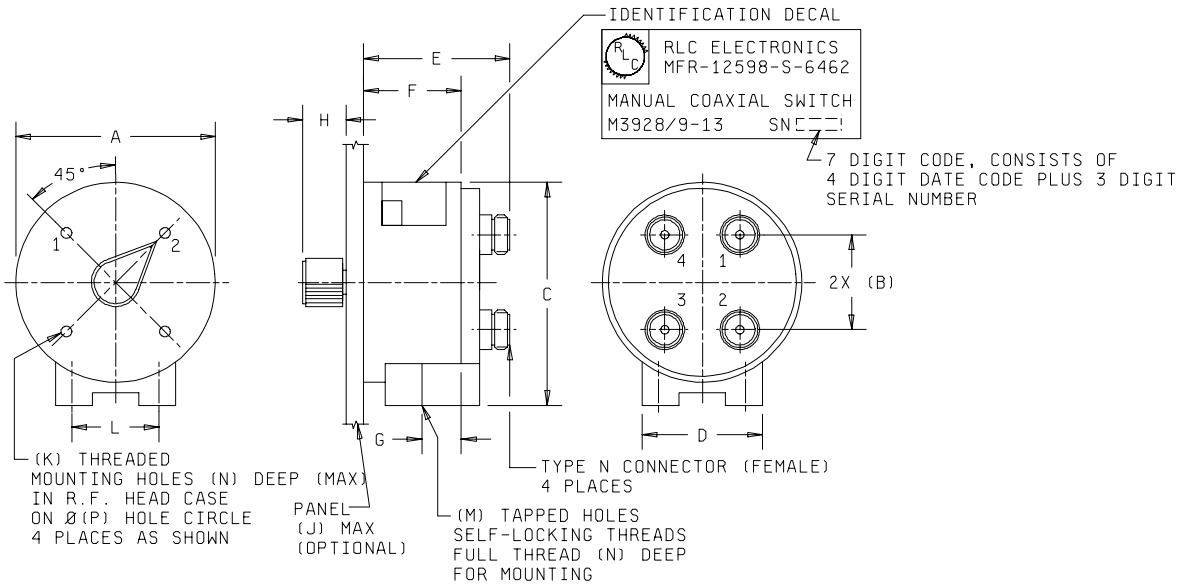
| Letter | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 1.197 | 1.177 | 30.40 | 29.90 |
| B | 1.135 | 1.115 | 28.83 | 28.32 |
| C | 1.84 | 1.78 | 46.7 | 45.2 |
| D | .807 | .787 | 20.5 | 20.0 |
| E | .190-32 UNF-2B | | 4.8-32 UNF-2B | |
| F | .322 | .302 | 8.18 | 7.67 |
| G | .250 | --- | 6.35 | --- |
| H | 3.260 | 3.240 | 82.80 | 82.30 |
| J | 1.78 | 1.72 | 45.2 | 43.7 |
| K | .112-40 UNC-2B | | 2.84-40 UNC-2B | |
| L | .25 | .19 | 6.4 | 4.8 |
| M | 2.232 | 2.212 | 56.69 | 56.18 |
| N | 1.822 | 1.802 | 46.28 | 45.77 |
| P | .822 | .802 | 20.88 | 20.37 |
| Q | 3.010 | 2.990 | 76.45 | 75.95 |

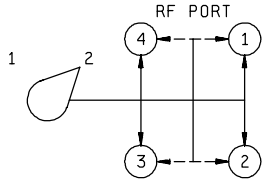
NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 inch (± 0.25 mm) for three place decimals and ± .03 inch (± 0.8 mm) for two place decimals.

FIGURE 1. Dimensions and configuration for PIN M3928/9-04 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.




POSITION 1 CONNECTS 1 TO 4 | 2 TO 3
POSITION 2 CONNECTS 1 TO 2 | 3 TO 4

SCHEMATIC

(Outline and schematic for M3928/9-13)

FIGURE 2. Dimensions and configurations for PINs M3928/9-05 and M3928/9-13.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

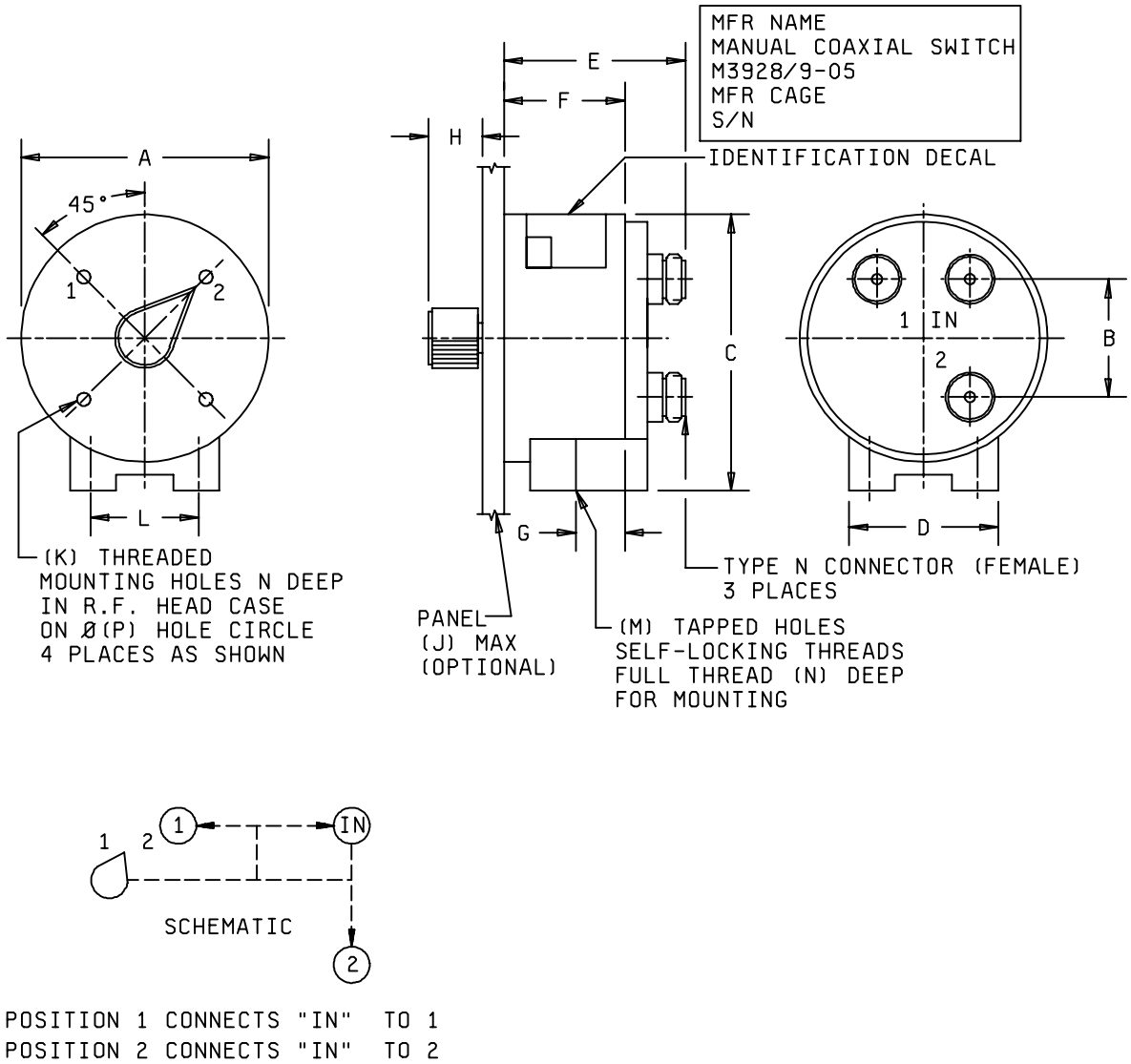
| Letter | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Max | Min | Max | Min |
| A | 2.760 dia | 2.740 dia | 70.10 dia | 69.60 dia |
| B | 1.072 | 1.052 | 27.23 | 26.72 |
| C | 2.970 | 2.950 | 75.44 | 74.93 |
| D | 1.822 | 1.802 | 46.28 | 45.77 |
| E | 2.300 | 2.280 | 58.42 | 57.91 |
| F | 1.463 | 1.443 | 37.16 | 36.65 |
| G | .572 | .552 | 14.53 | 14.02 |
| H | .854 | .834 | 21.69 | 21.18 |
| J | .250 | --- | 6.35 | --- |
| K | .112-40 UNC-2B | | 2.84-40 UNC-2B | |
| L | 1.187 Nom | | 30.15 Nom | |
| M | .190-32 UNF-2B | | 4.83-32 UNF-2B | |
| N | .312 | --- | 7.92 | --- |
| P | 2.232 dia | 2.212 dia | 56.69 dia | 56.18 dia |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 inch (± 0.25 mm) for three place decimals and ± .02 inch (± 0.5 mm) for two place decimals.
4. Remove all burrs and sharp edges .005 max. Finish to be .03 microinch or better.

FIGURE 2. Dimensions and configurations for PINs M3928/9-05 and M3928/9-13 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

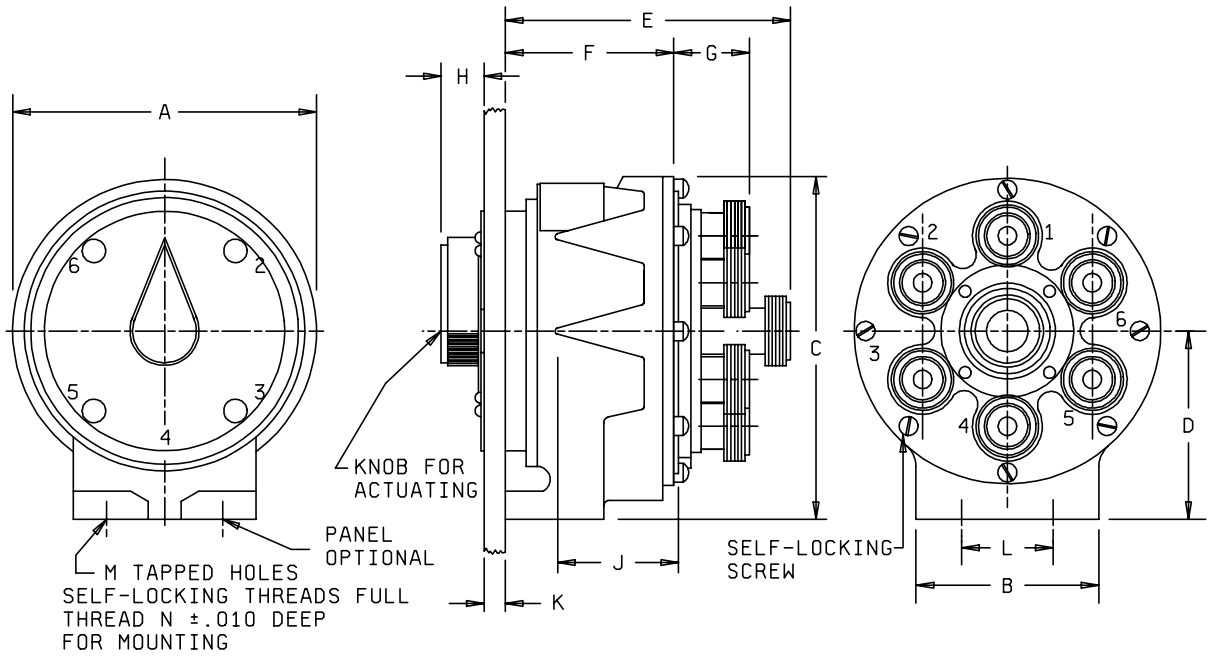



(Outline and schematic for M2928/9-05)

FIGURE 2. Dimensions and configurations for PINs M3928/9-05 and M3928/9-13 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.




| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | 3.010 dia | 2.990 dia | 76.45 dia | 75.95 dia | G | .822 | .802 | 20.88 | 20.37 |
| B | 1.84 | 1.78 | 46.7 | 45.2 | H | .807 | .787 | 20.50 | 19.99 |
| C | 3.260 | 3.240 | 82.80 | 82.30 | J | 1.135 | 1.115 | 28.83 | 28.32 |
| D | 1.78 | 1.72 | 45.2 | 43.7 | K | .250 | --- | 6.35 | --- |
| E | 3.104 | 3.084 | 78.84 | 78.33 | L | 1.197 | 1.177 | 30.40 | 29.90 |
| F | 1.822 | 1.802 | 46.28 | 45.77 | | | | | |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 inch (± 0.25 mm) for three place decimals and ± .03 inch (± 0.8 mm) for two place decimals.

FIGURE 3. Dimensions and configuration for PINs M3928/9-14, and M3928/9-15.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

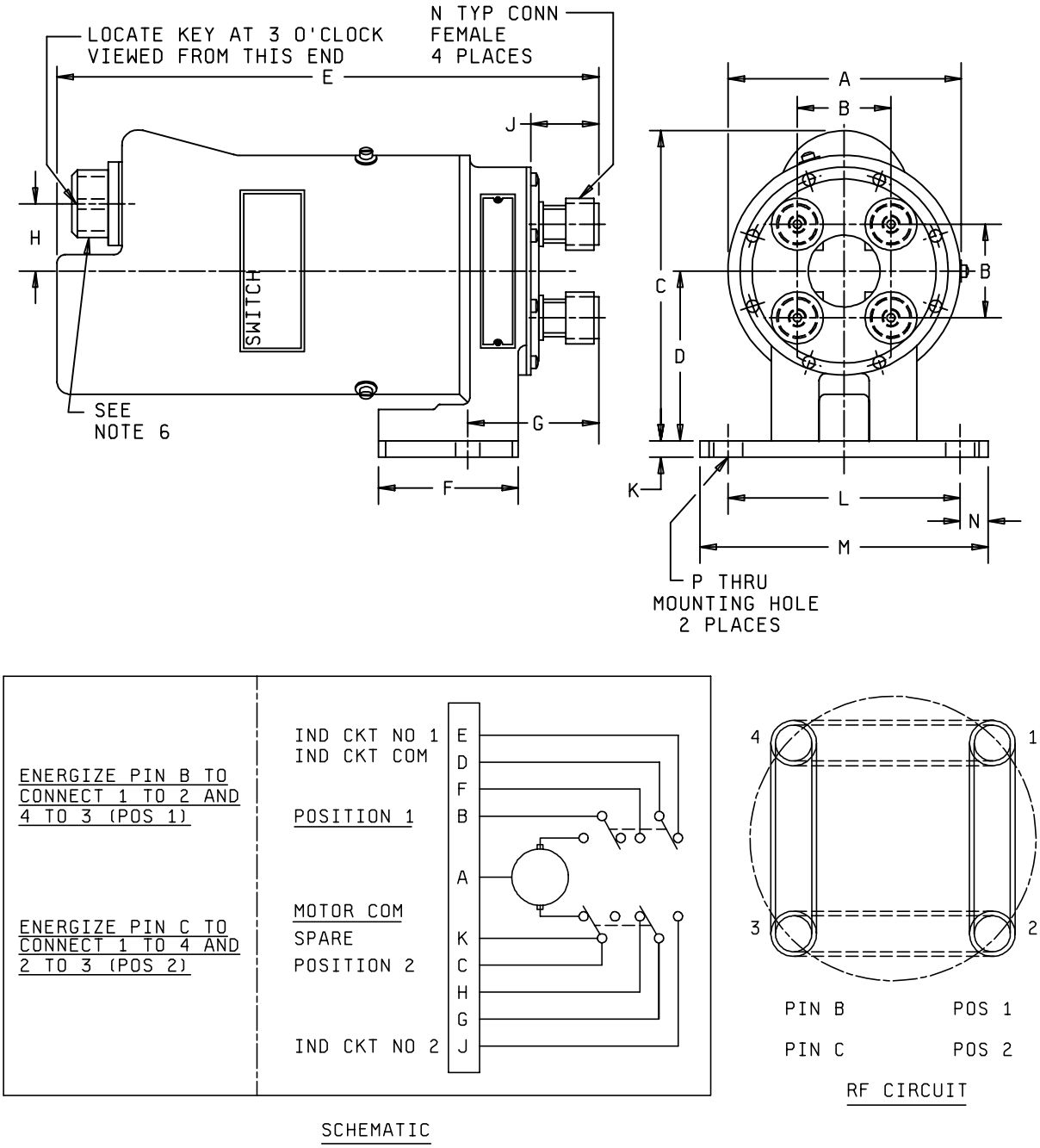



FIGURE 4. Dimensions and configuration for PIN M3928/9-17.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | 2.838 dia | 2.818 dia | 72.09 dia | 71.58 dia | H | .979 | .959 | 24.87 | 24.36 |
| B | 1.072 | 1.052 | 27.23 | 26.72 | J | .94 | .88 | 23.9 | 22.4 |
| C | 3.557 | 3.537 | 90.35 | 89.84 | K | .260 | .240 | 6.60 | 6.10 |
| D | 1.885 | 1.865 | 47.88 | 47.37 | L | 2.760 | 2.740 | 70.10 | 69.60 |
| E | 6.65 | --- | 168.9 | --- | M | 3.53 | 3.47 | 89.7 | 88.1 |
| F | 1.635 | 1.615 | 41.53 | 41.02 | N | .385 | .365 | 9.78 | 9.27 |
| G | 1.948 | 1.928 | 49.48 | 48.97 | P | .287 dia | .280 dia | 7.29 dia | 7.11 dia |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 inch (± 0.25 mm) for three place decimals and ± .03 inch (± 0.76 mm) for two place decimals.
4. Remove all burrs and sharp edges .015 inch (0.38 mm) maximum.
5. All machined surfaces shall be in accordance with ANSI 946.1.
6. Mates with MS3452 W-12-10S.
7. For PIN M3928/9-04, the center connector shall be deleted.

FIGURE 4. Dimensions and configuration for PIN M3928/9-17 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and performance characteristics.

| PIN M3928/9-1/ | Figure No. | Housing | Manual or remote | Solenoid or motor | Frequency range dc to GHz | VSWR | Insertion loss (dB) | Isolation (dB) | Configur-ation | Position indicating circuit (and current rating) | Life cycles x 1000 | Nominal operating voltage | Operating current (A) | Switch time (ms) | Power connector | Weight (lbs) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | max | max | min | | | | | max | | | max |
| 04N, S | 1 | I | M | --- | 10 | 1.3 | 0.31 | 50 | 2P2T | None | 200 | --- | --- | --- | --- | 1.75 |
| 05N, S | 2 | I | M | --- | 10 | 1.3 | 0.31 | 50 | 1P2T | None | 200 | --- | --- | --- | --- | 1.75 |
| 13N, S | 2 | I | M | --- | 10 | 1.3 | 0.31 | 50 | TR | None | 200 | --- | --- | --- | --- | 1.75 |
| 14N, S | 3 | I | M | --- | 10 | 1.3 | 0.31 | 50 | 1P3T | None | 200 | --- | --- | --- | --- | 1.75 |
| 15N, S | 3 | I | M | --- | 10 | 1.3 | 0.31 | 50 | 1P6T | None | 200 | --- | --- | --- | --- | 1.75 |
| 17N, S | 4 | I | R | M | 12.4 | 1.5 | 0.5 | 60 | TR | Yes | 60 | 28 V dc | 3.0 | 200 | See fig 3 | 3 |

1/ M3928/9-01, M3928/9-02, M3928/9-03, M3928/9-06, M3928/9-07, M3928/9-08, M3928/9-09, M3928/9-10, M3928/9-11, M3928/9-12, and M3928/9-16 have been deleted.





Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figures 1 through 4.

Electrical and performance characteristics: See table I.

RF power handling capability (average): 100 watts (min).

RF connectors: Series N connectors shall mate with series N male connectors in accordance with MIL-PRF-39012/1.

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55 deg C to +85 deg C.

Part or Identifying Number (PIN): M3928/9- (dash number from table I).

TABLE II. PIN to type cross-reference. 1/

| PIN M3928/9- | Type SA- /U |
|---|---|
| 04 | 1336 |
| 05 | 1337 |
| 13 | 1345 |
| 14 | 1346 |
| 15 | 1347 |
| 17 | --- |

1/ PINs M3928/9-01, M3928/9-02, M3928/9-03, M3928/9-06, M3928/9-07, M3928/9-08, M3928/9-09, M3928/9-10, M3928/9-11, M3928/9-12, and M3928/9-16 have been deleted.

Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - MC, OS
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5985-1209-03)



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.